

Question:

Install Burp Suite Community Edition

install:

- **Burp Suite Community Edition**
- **Firefox (recommended)**

Capture Your First Request

Use a test website or your own practice application.

- 1. Turn Intercept ON**
- 2. Visit a webpage**
- 3. Capture the request**
- 4. Observe:**
 - a. URL**
 - b. Method**
 - c. Headers**
 - d. Cookies**

Repeater

- 1. Send request to Repeater.**
- 2. Change harmless values such as:**
 - a. Search keyword**
 - b. Page number**
 - c. Language parameter**
- 3. Resend request.**
- 4. Observe response changes.**

Assignment

Choose a website you are authorized to test.

Capture:

- 1. Homepage Request**
- 2. Login Page Request**
- 3. Search Request**

For each request explain:

- **Method (GET/POST)**
- **URL**
- **Headers**
- **Parameters**
- **Response Code**

Take screenshots and prepare a report.

Objective

The objective of this practical was to learn how to intercept, analyze, and modify HTTP requests and responses using Burp Suite Community Edition with the OWASP Juice Shop application.

Tools Used

1. Burp Suite Community Edition
2. Mozilla Firefox
3. OWASP Juice Shop

Website Used

OWASP Juice Shop Demo Application

Website:

<https://juice-shop.herokuapp.com/>

Procedure

Step 1: Installation of Burp Suite

Burp Suite Community Edition was downloaded and installed successfully on the Windows operating system.

Step 2: Installation of Mozilla Firefox

Mozilla Firefox browser was installed and used for web traffic interception through Burp Suite.

Step 3: Proxy Configuration

Firefox proxy settings were configured manually using:

- HTTP Proxy: 127.0.0.1
- Port: 8080

This allowed all browser traffic to pass through Burp Suite.

Step 4: Opening OWASP Juice Shop

The OWASP Juice Shop application was opened in Firefox browser.

Burp Suite successfully intercepted and recorded the website requests.

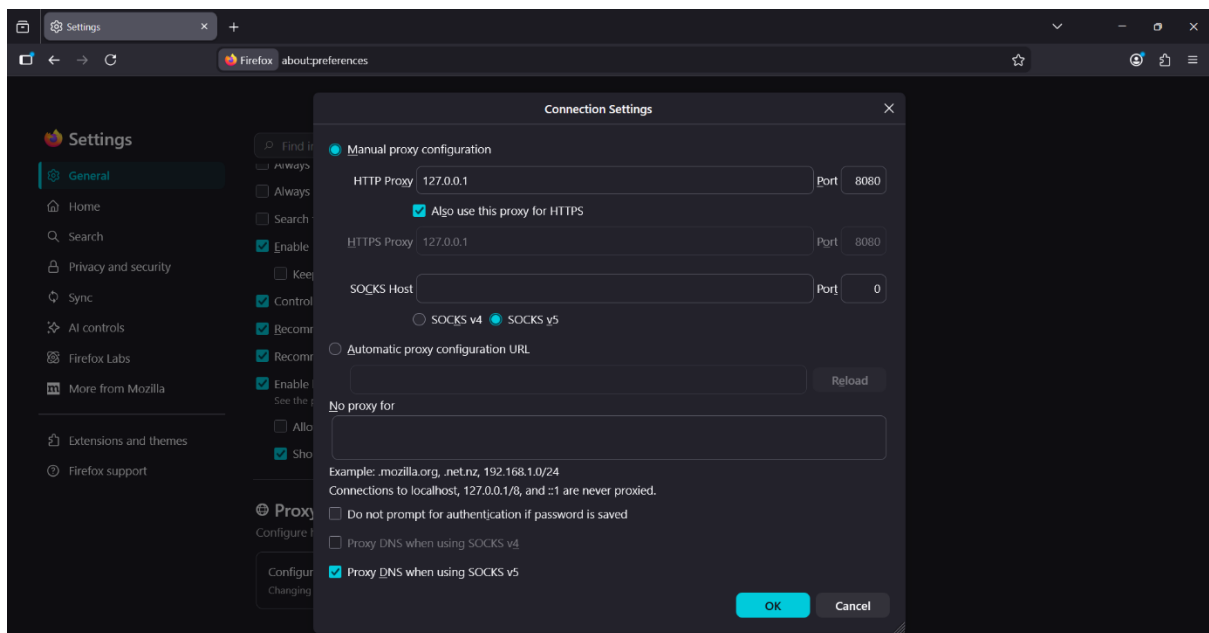
Firefox Proxy Configuration

Firefox browser proxy settings were configured manually to redirect all traffic through Burp Suite.

Proxy Details:

- IP Address: 127.0.0.1
- Port: 8080

Screenshot : Firefox Proxy Settings



Homepage Request Analysis

Method:-GET

URL:-<http://juice-shop.herokuapp.com/>

Headers Observed

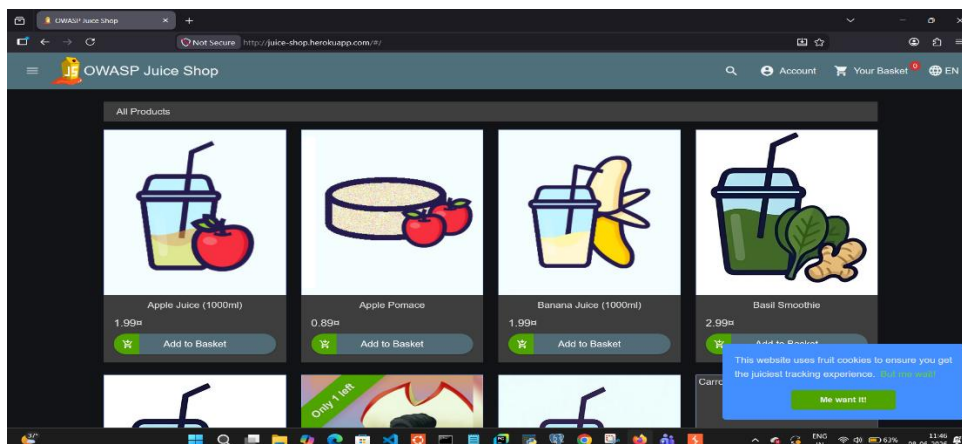
- Host
- User-Agent
- Accept
- Accept-Language
- Connection

Response Code:-200

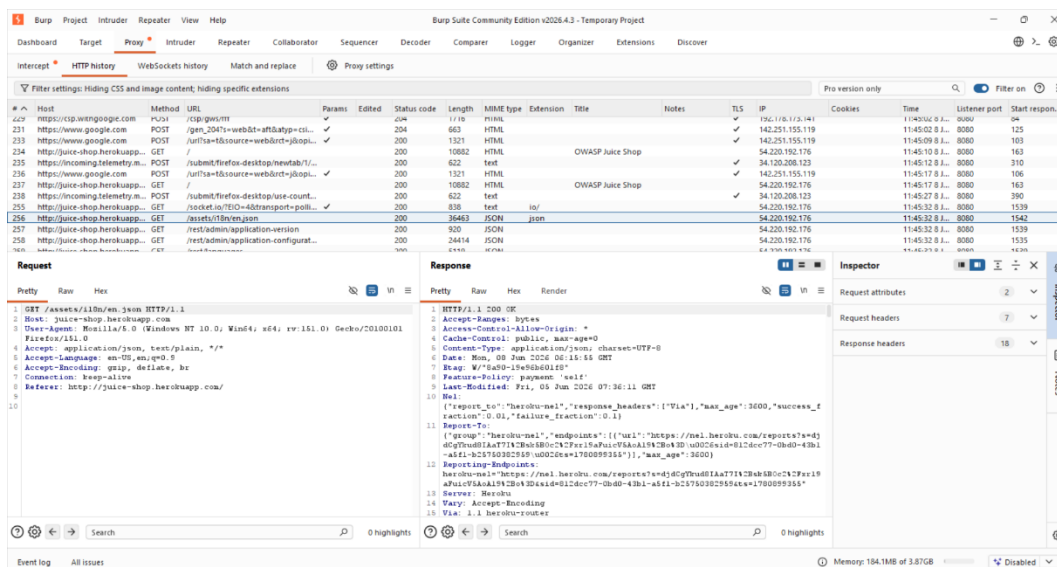
Observation

The homepage request was successfully intercepted through Burp Suite. The request used the GET method to retrieve the main page of the application.

Screenshot: OWASP Juice Shop Homepage



Screenshot : Burp HTTP History



Screenshot : Homepage Request Capture

Request

PrettyRawHex

GET / HTTP/1.1

Host: juice-shop.herokuapp.com

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101 Firefox/151.0

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Language: en-US,en;q=0.9

Accept-Encoding: gzip, deflate, br

Referer: https://www.google.com/

Connection: keep-alive

Upgrade-Insecure-Requests: 1

Priority: u=0, i

Response

PrettyRawHexRender

1 HTTP/1.1 200 OK

2 Accept-Ranges: bytes

3 Access-Control-Allow-Origin: *

4 Cache-Control: public, max-age=0

5 Content-Type: text/html; charset=UTF-8

6 Date: Mon, 08 Jun 2026 06:15:27 GMT

7 Etag: W/"26af-19ea5946414"

8 Feature-Policy: payment 'self'

9 Last-Modified: Mon, 08 Jun 2026 04:53:46 GMT

10 Nel: {

11 "report_to": "heroku-nel", "response_headers": ["Via"], "max_age": 3600, "success_fr

12 action": 0.01, "failure_fraction": 0.1}

13 Report-To: {

14 "group": "heroku-nel", "endpoints": [{"url": "https://nel.heroku.com/reports?s=F42

15 BGvglpH70mcWgIVhVs5t2FBujdgOHw892v0bx4oUvL4s43D\u0026sid=812dce77-0bd0-43b1-a5f

16 1-b25750382959\u0026ets=1780899327"}], "max_age": 3600}

17 Reporting-Endpoints:

18 heroku-nel="https://nel.heroku.com/reports?s=F42BGvglpH70mcWgIVhVs5t2FBujdgOHw8

19 92vDbx4oUvL4s43D&sid=812dce77-0bd0-43b1-a5f1-b25750382959&ets=1780899327"

20 Server: Heroku

21 Vary: Accept-Encoding

22 Via: 1.1 heroku-router

Inspector

Request attributes2

Request headers9

Response headers18

Inspector

Notes

Event logAll issues

Memory: 179.0MB of 3.87GB

Disabled

Login Request Analysis

Method:-POST

URL:-/rest/user/login

Parameters

- email = test@gmail.com
- password = test123

Headers Observed

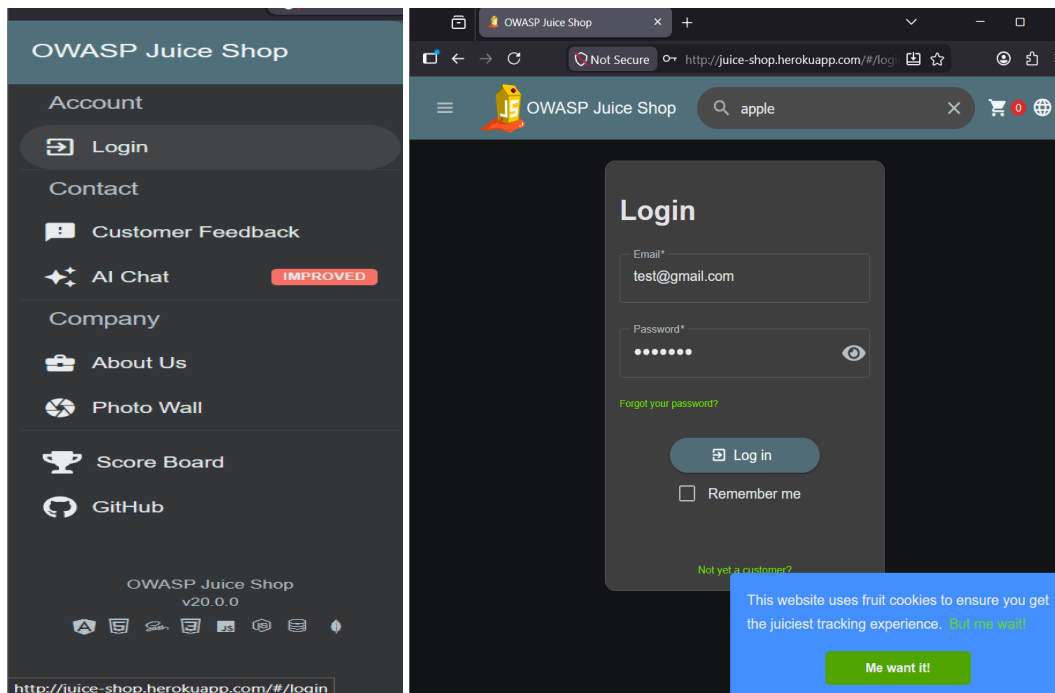
- Content-Type
- User-Agent
- Cookie
- Origin

Response Code:-200

Observation

The login request used the POST method to securely send user credentials to the server. The request and response were successfully captured and analysed in Burp Suite.

Screenshot : OWASP Juice Shop Login page



Screenshot 5: Login page Request Capture

POST request to http://juice-shop.herokuapp.com/rest/user/login

Request **Response**

Pretty Raw Hex

```
1 POST /rest/user/login HTTP/1.1
2 Host: juice-shop.herokuapp.com
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101 Firefox/151.0
4 Accept: application/json, text/plain, */*
5 Accept-Language: en-US,en;q=0.9
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/json
8 Content-Length: 47
9 Origin: http://juice-shop.herokuapp.com
10 Connection: keep-alive
11 Referer: http://juice-shop.herokuapp.com/
12 Cookie: language=en; welcomebanner_status=dismiss
13 Priority: u=0
14
15 {
  "email": "test@gmail.com",
  "password": "test123"
}
```

POST request to http://juice-shop.herokuapp.com/rest/user/login

Request **Response**

Pretty Raw Hex Render

```
1 HTTP/1.1 401 Unauthorized
2 Access-Control-Allow-Origin: *
3 Content-Length: 26
4 Content-Type: text/html; charset=utf-8
5 Date: Mon, 08 Jun 2026 06:27:25 GMT
6 Etag: W/"la-Ak3vVKfsmzAF3QQue2mDSG+3Bus"
7 Feature-Policy: payment 'self'
8 Hel: ("report_to":"heroku-nel","response_headers":{"Via"},"max_age":3600,"success_fraction":0.01,"failure_fraction":0.1)
9 Report-To:
  ("group":"heroku-nel","endpoints":[{"url":"https://nel.heroku.com/reports?s=0L42BHaSEzvLlNo0yNHD<3DF2ylut2B42BNrcR7sjle4KV6Bo43D\u0026sid=812dcc77-0bd0-43b1-a5f1-b25750382959&ts=1780900045"}],"max_age":3600)
10 Reporting-Endpoints:
  heroku-nel="https://nel.heroku.com/reports?s=0L42BHaSEzvLlNo0yNHD<3DF2ylut2B42BNrcR7sjle4KV6Bo43D\u0026sid=812dcc77-0bd0-43b1-a5f1-b25750382959&ts=1780900045"
11 Server: Heroku
12 Vary: Accept-Encoding
13 Via: 1.1 heroku-router
14 X-Content-Type-Options: nosniff
15 X-Frame-Options: SAMEORIGIN
16 X-Recruiting: /#/jobs
17
18 Invalid email or password.
```

Inspector

Request headers 12

Name	Value	
Host	juice-shop.herokuapp.c...	>
User-Agent	Mozilla/5.0 (Windows NT...	>
Accept	application/json, text/pl...	>
Accept-Language	en-US,en;q=0.9	>
Accept-Encoding	gzip, deflate, br	>
Content-Type	application/json	>
Content-Length	47	>
Origin	http://juice-shop.heroku...	>
Connection	keep-alive	>
Referer	http://juice-shop.heroku...	>
Cookie	language=en; welcomeb...	>
Priority	u=0	>

Response headers			15	^
Name	Value			
Access-Control-Allow-Or...	*			>
Content-Length	26			>
Content-Type	text/html; charset=utf-8			>
Date	Mon, 08 Jun 2026 06:27:...			>
Etag	W/"1a-ARJvVK+smzAF3Q...			>
Feature-Policy	payment 'self'			>
Nel	{"report_to":"heroku-nel"...			>
Report-To	{"group":"heroku-nel","e...			>
Reporting-Endpoints	heroku-nel="https://nel....			>
Server	Heroku			>
Vary	Accept-Encoding			>
Via	1.1 heroku-router			>
X-Content-Type-Options	nosniff			>

Inspector

Request attributes2^

Protocol

HTTP/1HTTP/2

Name	Value	
Method	POST	>
Path	/rest/user/login	>

Request cookies2^

Name	Value	
language	en	>
welcomebanner_status	dismiss	>

Search Request Analysis

Method:-POST

URL:-/socket.io/?EIO=4&transport=polling

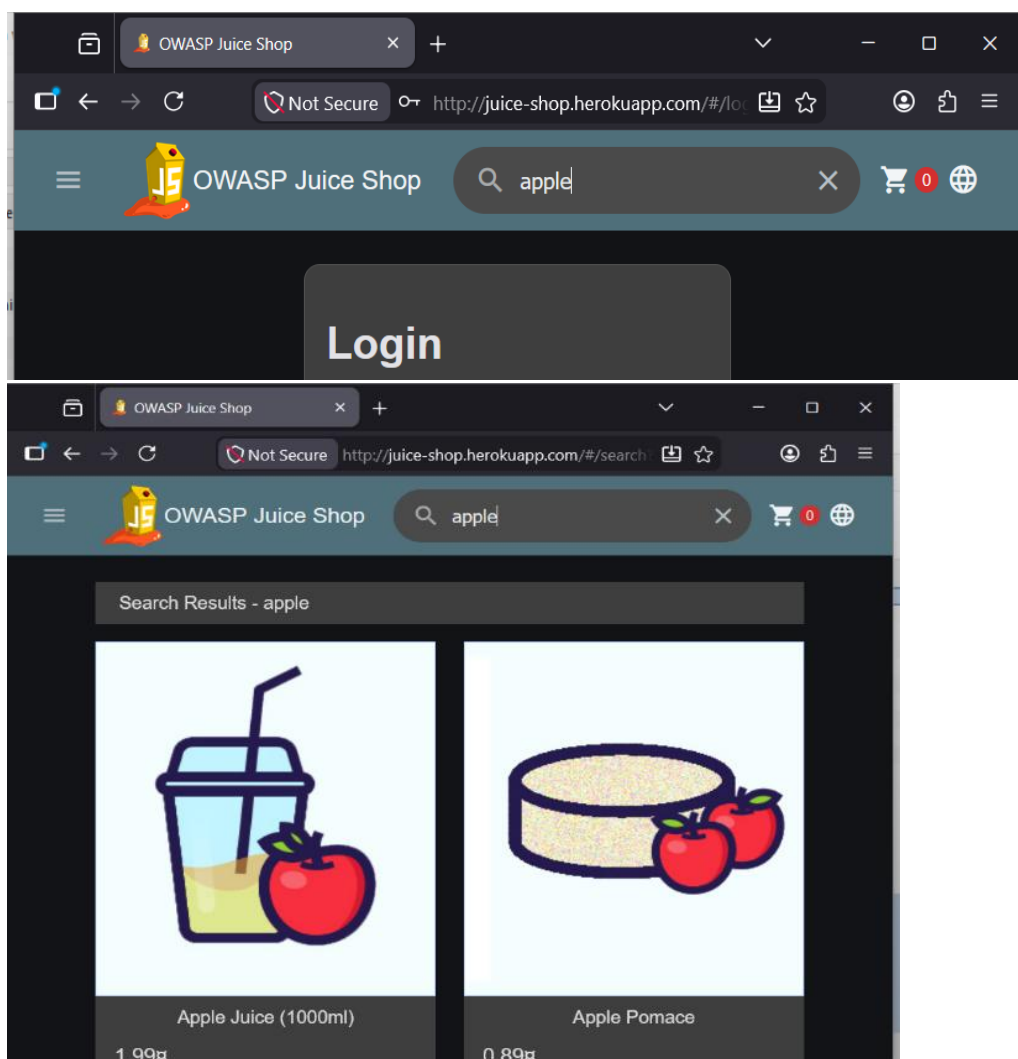
Parameters:-apple

Response Code:-200

Observation

The product search request was intercepted successfully. The keyword “apple” was transmitted to the server and related product information was returned in the response.

Screenshot : OWASP Juice Shop Search



Screenshot : Search Request Capture

⚡ Burp Project Intruder Repeater View Help

Burp Suite Community Edition v2026.4.3 - Temporary Project

Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Discover

Intercept HTTP history WebSockets history Match and replace Proxy settings

Filter settings: Hiding CSS and image content; hiding specific extensions

Pro version only Filter on

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start respon
179	https://prod-images.merino...	GET	/favicons/f3e579097b6a298dc549...		✓	304	213		oct			✓	34.111.211.209		12:22:31.8 J...	8080	67
181	https://mozilla-chttp.factly-e...	POST	/		✓	200	765					✓	151.101.209.91		12:22:34.8 J...	8080	322
182	http://juice-shop.herokuapp...	POST	/socket.io/?EIO=4&transport=polli...		✓	200	719	HTML	io/				46.137.15.86		12:22:34.8 J...	8080	179
183	http://juice-shop.herokuapp...	GET	/socket.io/?EIO=4&transport=polli...		✓	200	766	text	io/				46.137.15.86		12:22:34.8 J...	8080	179
184	http://juice-shop.herokuapp...	GET	/socket.io/?EIO=4&transport=web...		✓	101	717		io/				46.137.15.86		12:22:34.8 J...	8080	180
104	http://juice-shop.herokuapp...	GET	/api/Quantity/		✓	304	810						46.137.15.86		12:22:49.8 J...	8080	189
105	http://juice-shop.herokuapp...	GET	/test/products/search?q=		✓	304	810						46.137.15.86		12:22:49.8 J...	8080	179
106	https://incoming.telemetry...	POST	/submit/ffox-desktop/baseline/...		✓							✓	34.120.200.123		12:22:52.8 J...	8080	
107	http://juice-shop.herokuapp...	POST	/socket.io/?EIO=4&transport=polli...		✓				io/				46.137.15.86		12:22:52.8 J...	8080	
108	http://juice-shop.herokuapp...	GET	/socket.io/?EIO=4&transport=polli...		✓				io/				46.137.15.86		12:22:52.8 J...	8080	
110	http://juice-shop.herokuapp...	POST	/socket.io/?EIO=4&transport=polli...		✓				io/				46.137.15.86		12:23:05.8 J...	8080	
111	http://juice-shop.herokuapp...	GET	/socket.io/?EIO=4&transport=polli...		✓				io/				46.137.15.86		12:23:07.8 J...	8080	

Request

Pretty Raw Hex

```
1 POST /socket.io/?EIO=4&transport=polling&EIO=4&transport=polling&EIO=4&transport=polling HTTP/1.1
2 Host: juice-shop.herokuapp.com
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101 Firefox/151.0
4 Accept: */*
5 Accept-Language: en-US,en;q=0.9
6 Accept-Encoding: gzip, deflate, br
7 Content-type: text/plain;charset=UTF-8
8 Content-length: 27
9 Origin: http://juice-shop.herokuapp.com
10 Connection: keep-alive
11 Referer: http://juice-shop.herokuapp.com/
12 Cookie: language=en; welcomebanner_status=dismiss
13
14 [{"verifyLocalXssChallenge","apple"}]
```

Inspector

Request query parameters 4

Request body parameters 1

Request cookies 2

Request headers 11

Name	Value
language	en
welcomebanner_status	dismiss

Name	Value
Host	juice-shop.herokuapp...
User-Agent	Mozilla/5.0 (Windows...
Accept	*/*
Accept-Language	en-US,en;q=0.9
Accept-Encoding	gzip, deflate, br
Content-type	text/plain;charset=UTF...

Search

0 highlights

Found 100 hits

100 hits

Memory: 100.7MB of 1.87GB

Disabled

Repeater Analysis

The search request was sent to Burp Repeater for testing.

The original parameter:apple

was modified to:banana

The modified request was resent to the server using the Repeater tool.

After modification, the server response changed according to the new keyword value.

Modified Parameter

apple → banana

Observation

Burp Repeater successfully allowed modification and resending of the request. The server responded differently after changing the request parameter.

Screenshot : Repeater Modified Request

The screenshot displays the Burp Suite interface. The top panel shows a list of HTTP history items. Item 507 is selected, showing a POST request to `http://juice-shop.herokuapp.com/0Mwj&sid=6-ttoYWiRHoojf4JACQZ`. The bottom panel shows the 'Request' tab in the Repeater tool. The request is displayed in 'Pretty' format, showing a POST to `/socket.io/?EIO=4&transport=polling` with various headers and a body containing a JSON array: `42[{"verifyLocalXssChallenge", "apple"}]`. The 'Send' button is visible above the request details.

No.	URL	Method	Status	Size	Content-Type
481	https://mozilla-ohhttp.fastly-e...	POST	/	200	765
482	http://juice-shop.herokuapp.com/...	POST	/socket.io/?EIO=4&transport=polli...	200	719
483	http://juice-shop.herokuapp.com/...	GET	/socket.io/?EIO=4&transport=polli...	200	766
484	http://juice-shop.herokuapp.com/...	GET	/socket.io/?EIO=4&transport=web...	101	717
504	http://juice-shop.herokuapp.com/...	GET	/api/Quantities/	304	810
505	http://juice-shop.herokuapp.com/...	GET	/rest/products/search?q=	304	810
506	https://incoming.telemetry.m...	POST	/submit/firefox-desktop/baseline/1...		
507	http://juice-shop.herokuapp.com/...	POST	/socket.io/?EIO=4&transport=polli...		
508	http://juice-shop.herokuapp.com/...	GET	/socket.io/?EIO=4&transport=polli...		
510	http://juice-shop.herokuapp.com/...	POST	/socket.io/?EIO=4&transport=polli...		
511	http://juice-shop.herokuapp.com/...	GET	/socket.io/?EIO=4&transport=polli...		
512	http://juice-shop.herokuapp.com/...	GET	/socket.io/?EIO=4&transport=polli...		

Request

Pretty Raw Hex

```
1 POST /socket.io/?EIO=4&transport=polling HTTP/1.1
2 Host: juice-shop.herokuapp.com
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101 Firefox/151.0
4 Accept: */*
5 Accept-Language: en-US,en;q=0.9
6 Accept-Encoding: gzip, deflate, br
7 Content-type: text/plain;charset=UTF-8
8 Content-length: 37
9 Origin: http://juice-shop.herokuapp.com
10 Connection: keep-alive
11 Referer: http://juice-shop.herokuapp.com/
12 Cookie: language=en; welcomebanner_status=dismiss
13
14 42[{"verifyLocalXssChallenge", "apple"}]
```

⚡ Burp Project Intruder Repeater View Help

Dashboard Target Proxy **Intruder** Repeater Collaborator Sequencer Deco

3 x +

Send Cancel < > ⚡ Burp AI

Request

Pretty Raw Hex

```
1 POST /socket.io/?EIO=4&transport=polling&t=Pwc0Mwj&sid=6-ttoYWiRHooj4JACQZ
2 HTTP/1.1
3 Host: juice-shop.herokuapp.com
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101
5 Firefox/151.0
6 Accept: */*
7 Accept-Language: en-US,en;q=0.9
8 Accept-Encoding: gzip, deflate, br
9 Content-type: text/plain;charset=UTF-8
10 Content-Length: 40
11 Origin: http://juice-shop.herokuapp.com
12 Connection: keep-alive
13 Referer: http://juice-shop.herokuapp.com/
14 Cookie: language=en; welcomebanner_status=dismiss
15 42["verifyLocalXssChallenge","banana"]
```

3 x +

Send Cancel < > ⚡ Burp AI

Target: http://juice-shop.herokuapp.com HTTP/1

Request

Pretty Raw Hex

```
1 POST /socket.io/?EIO=4&transport=polling&t=Pwc0Mwj&sid=6-ttoYWiRHooj4JACQZ
2 HTTP/1.1
3 Host: juice-shop.herokuapp.com
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101
5 Firefox/151.0
6 Accept: */*
7 Accept-Language: en-US,en;q=0.9
8 Accept-Encoding: gzip, deflate, br
9 Content-type: text/plain;charset=UTF-8
10 Content-Length: 40
11 Origin: http://juice-shop.herokuapp.com
12 Connection: keep-alive
13 Referer: http://juice-shop.herokuapp.com/
14 Cookie: language=en; welcomebanner_status=dismiss
15 42["verifyLocalXssChallenge","banana"]
16
```

Response

Pretty Raw Hex Render

```
1 HTTP/1.1 400 Bad Request
2 Access-Control-Allow-Origin: http://localhost:4200
3 Content-Type: application/json
4 Date: Mon, 09 Jun 2026 07:05:10 GMT
5 Nel:
6 {"report_to":"heroku-nel","response_headers":["Via"],"max_age":3600,"success_fr
7 action":0.01,"failure_fraction":0.1}
8 Report-To:
9 {"group":"heroku-nel","endpoints":[{"url":"https://nel.heroku.com/reports?s=HT7
10 7No4hJcmAb2KoPCQeep0CTq3k2FCagk1x50nBhavY43D-u0026sid=812dcc77-0bd0-43b1-a5f1-
11 b25750382559-u0026ts=1780902310"}],"max_age":3600}
12 Reporting-Endpoints:
13 heroku-nel="https://nel.heroku.com/reports?s=HT77No4hJcmAb2KoPCQeep0CTq3k2FCag
14 k1x50nBhavY43D-u0026sid=812dcc77-0bd0-43b1-a5f1-b25750382559&ts=1780902310"
15 Server: Heroku
16 Vary: Origin
17 Via: 1.1 heroku-router
18 Content-Length: 41
19 {
20   "code":1,
21   "message":"Session ID unknown"
22 }
```

Inspector

Request attributes 2

Request query parameters 4

Request body parameters 1

Request cookies 2

Request headers 11

Response headers 10